

Technical Datasheet Input / Output Modules with Modbus RTU Protocol with RS485 Interface

The IO module communicates via Ethernet Interface. The module is designed to allow interface with other Modbus TCP/IP compatible devices. The module acts as an input/output device between the Modbus TCP/IP Network and the Gateway.

Digital IO module is sturdy, low power usage and easy to use.

4 Port DO + 4 DI + 4 AI Module with Modbus TCP Ethernet: -

The IO modules enclosures are DIN rail Mountable. It's a closed Enclosure with



openings for connectors and LED indicators.

The design of the modules incorporates '**resettable Fuses**' to safeguard against reverse polarity connection for **Power Input**.

There are LED indicators for each of the IO ports as well as for the module Power.

Specifications

General –

I/O Connectors	22 Pin 5.08 mm pitch pluggable screw Terminal Block, 2 Pin 5.08 mm pitch pluggable screw terminals.
Dimensions	110 mm L x 110 mm B x 50 mm H
Power	Input Power – 12-24 V DC Typical – 12V DC @ 120mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



DO with Relay Output –

Channels	4
Contact Form	1A, 1C
Contact Material	Ag Alloy
Contact Capacity	10A @ 240VAC, 10A @ 28VDC
Coil Voltage	5 – 48 VDC
Coil Power	0.36 W
Insulation Resistance	250MΩ
Electrical Life	1 x 10 ⁵
Mechanical Life	1 x 10 ⁷
Operating Time	7 msec, Max 15 msec
Release Time	2 msec, Max 6 msec

wittelB™
 Our focus is You

DI Inputs –

Channels	4
Sense Voltage	3.3 – 30 VDC
Sense Logic	Logic '0' = <1 VDC, Logic '1' = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec

AI Inputs –

Channels	4
Input Signal	4 – 20 mA / 0 – 10VDC / NTC Type 3 (jumper selectable)
Accuracy	± 2% Full scale
Input Resolution	10 bits / 12 bits (Optional)
Isolation	Optically Isolated
External Loop Voltage	+ 12 VDC min
AI input impedance	120Ω

Configuration Settings: -

Communication Speed	10/100 Base-T Ethernet Interface
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	Configurable over Web Interface
Function code DO	0x05 and 0x0F (5 Single coil & 15 multiple coil)
DO Register Address	4,5,6,7.
Function code DI	0x02 Read discrete Input
DI Register Address	0,1,2,3.
Function Code AI	0x03 Read Holding Registers
AI Register Address	10 Bit - 2,3,4,5 / 12 Bit – 6,7,8,9.

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	DO 1	00001	0X05,0X0F	RS485	1 Bit Boolean
2	DO 2	00002	0X05,0X0F	RS485	1 Bit Boolean
3	DO 3	00003	0X05,0X0F	RS485	1 Bit Boolean
4	DO 4	00004	0X05,0X0F	RS485	1 Bit Boolean
5	DI 1(With & Without Potential)	10005	0X02	RS485	1 Bit Boolean
6	DI 2(With & Without Potential)	10006	0X02	RS485	1 Bit Boolean
7	DI 3(With & Without Potential)	10007	0X02	RS485	1 Bit Boolean
8	DI 4(With & Without Potential)	10008	0X02	RS485	1 Bit Boolean
9	AI 1 – 10 Bit	40002	0X03	RS485	16 Bit Unsigned int
10	AI 2 – 10 Bit	40003	0X03	RS485	16 Bit Unsigned int
11	AI 3 – 10 Bit	40004	0X03	RS485	16 Bit Unsigned int
12	AI 4 – 10 Bit	40005	0X03	RS485	16 Bit Unsigned int
13	AI 1 – 12 Bit	40006	0x03	RS485	16 Bit Unsigned int
14	AI 2 – 12 Bit	40007	0x03	RS485	16 Bit Unsigned int
15	AI 3 – 12 Bit	40008	0x03	RS485	16 Bit Unsigned int
16	AI 4 – 12 Bit	40009	0x03	RS485	16 Bit Unsigned int

Configuration Guide – Ethernet Modbus TCP/IP IO Module:-

1. Open a web browser (Google Chrome, Microsoft Edge, Firefox, etc.).
2. In the browser's address bar, type the default IP address:
192.168.0.160
3. Press **Enter**. A message will appear saying:
"Please go to <http://192.168.0.160/setup>."
4. Copy this link (<http://192.168.0.160/setup>) and paste it into a new browser tab.
5. Press **Enter** again to open the **Setup / Configuration Page** of the device.

To Reset the Device to Default Settings

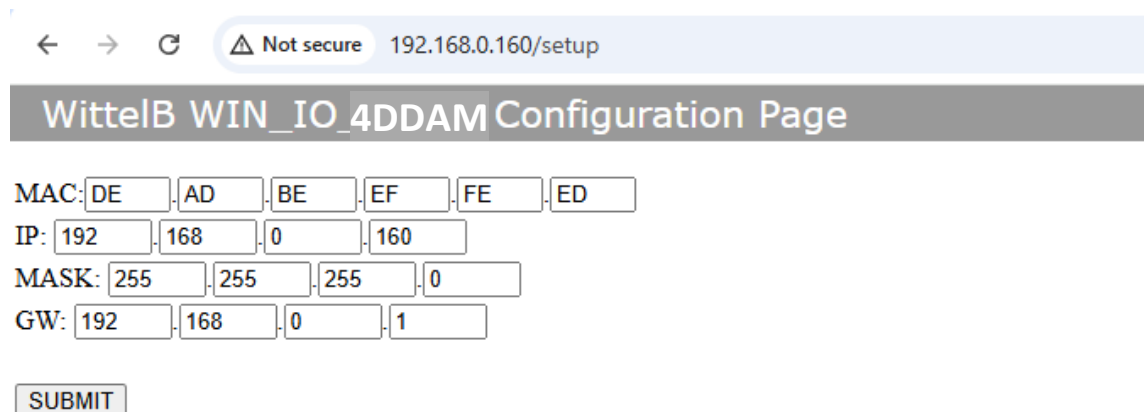
- Place the **jumper** on the designated **Reset position** on the device.
- Press the **Reset button** once.
- The module will reboot and revert to its default IP address: 192.168.0.160.

To Change the IP Address

- On the **Setup / Configuration Page**, enter the new IP address you wish to assign.
- Click **Submit** to save the new settings.
- The device will automatically apply the new IP address.

Note:

- Ensure your PC's IP address is in the **same network range** (for example, 192.168.0.xxx) when accessing the module.
- After changing the IP, you must use the **new address** to log in again.

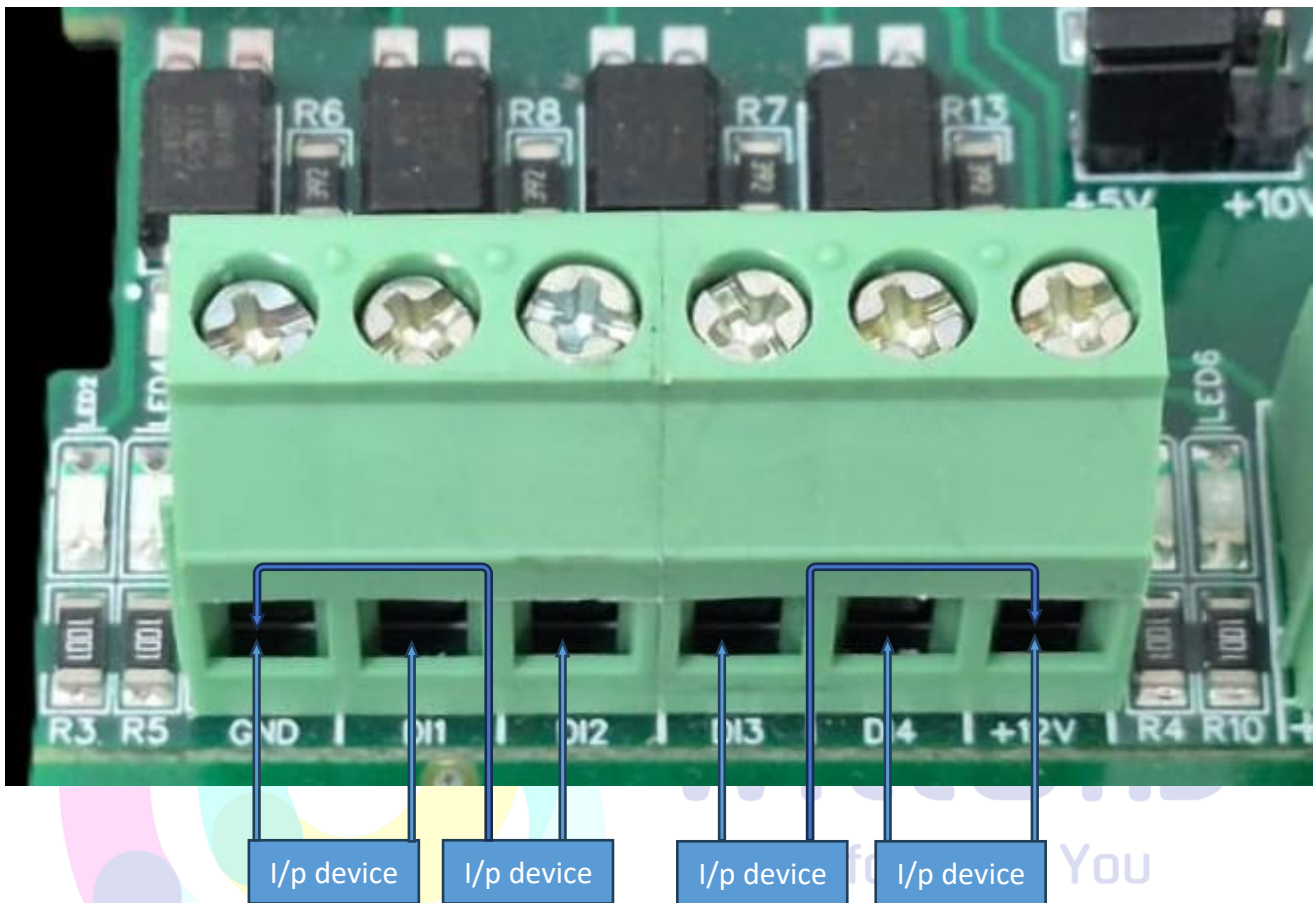


The screenshot shows a web browser window with the address bar displaying "192.168.0.160/setup". The page title is "WittelB WIN_IO_4DDAM Configuration Page". The form contains the following fields:

MAC:	DE	AD	BE	EF	FE	ED
IP:	192	168	0	160		
MASK:	255	255	255	0		
GW:	192	168	0	1		

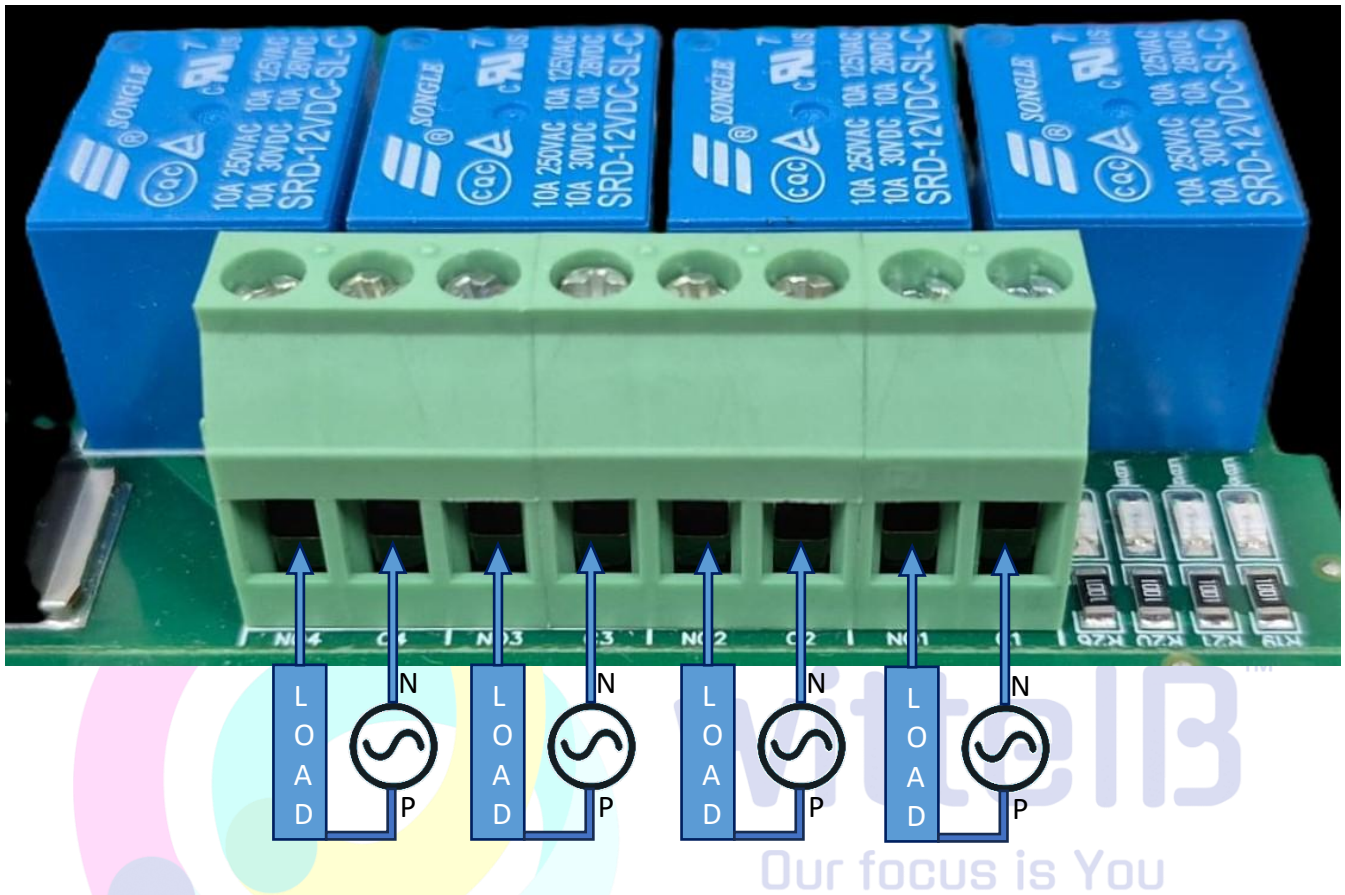
Below the fields is a **SUBMIT** button.

DI



- **Operation:** The module provides four isolated input channels (**DI1–DI4**) for sensing external signals or dry contacts. These inputs are opto-isolated for circuit protection.
- **Terminal Layout:**
 - **GND:** Common Ground reference.
 - **DI1 – DI4:** Individual Digital Input channels.
 - **+12V:** Internal reference voltage for sourcing "Dry Contact" switches.
- **Wiring Options:**
 - **Dry Contact:** Connect a switch between the **+12V** terminal and any **DI** pin.
 - **External Voltage:** Connect the external signal to a **DI** pin and the external ground to the **GND** terminal.
- **Voltage Rating:** The input circuitry is rated for a maximum of **80VDC**. Ensure external signals do not exceed this limit to prevent damage to the isolation ICs.

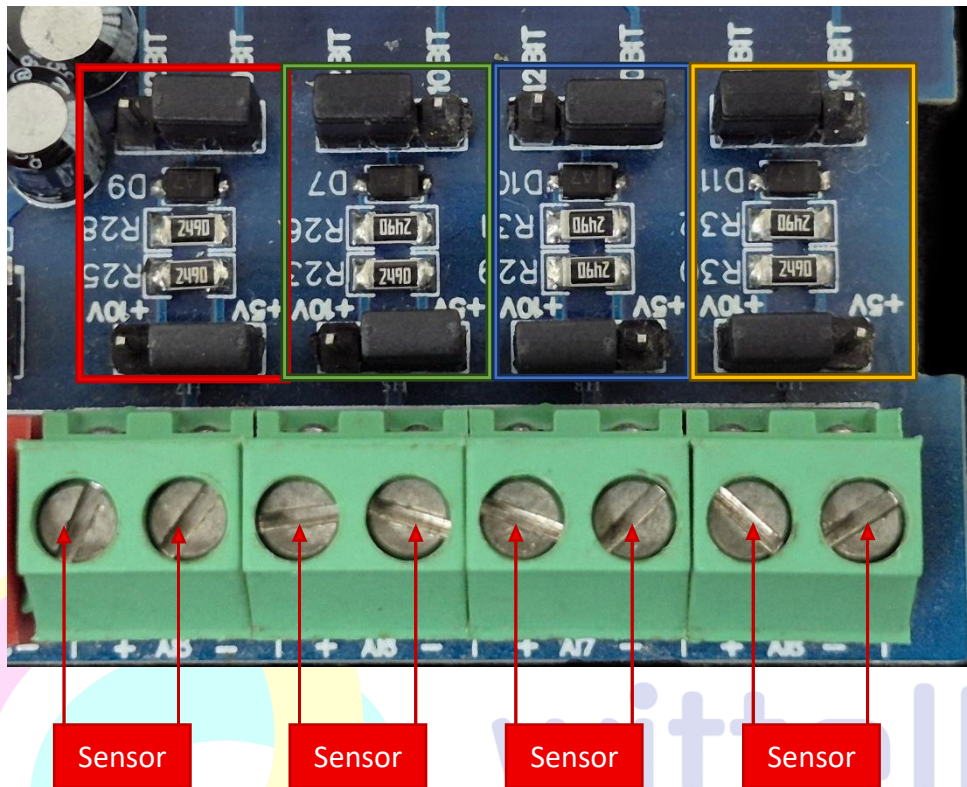
DO



- **Digital Output (DO) operation:** Each channel provides two terminals: **C (Common)** and **NO (Normally Open)**. When the relay coil is energized, terminal **C** is **internally connected to NO**, allowing current to flow. When the relay is de-energized, the connection between **C and NO** remains **open**, and the load stays OFF.
- **Load Wiring Connections:**
 - **AC Load Connection:** The **AC Live (Phase)** wire should be connected to terminal **C**. The **load is connected between terminal NO and Neutral (N)**.
 - **DC Load Connection:** The **positive supply (+V)** should be connected to terminal **C**. The **load is connected between terminal NO and Ground (GND)**.
- **Crucial Safety Recommendation and Rating:** It is strongly recommended to switch the Live (Phase) conductor in AC applications to ensure the load is safely de-energized when the relay is off.

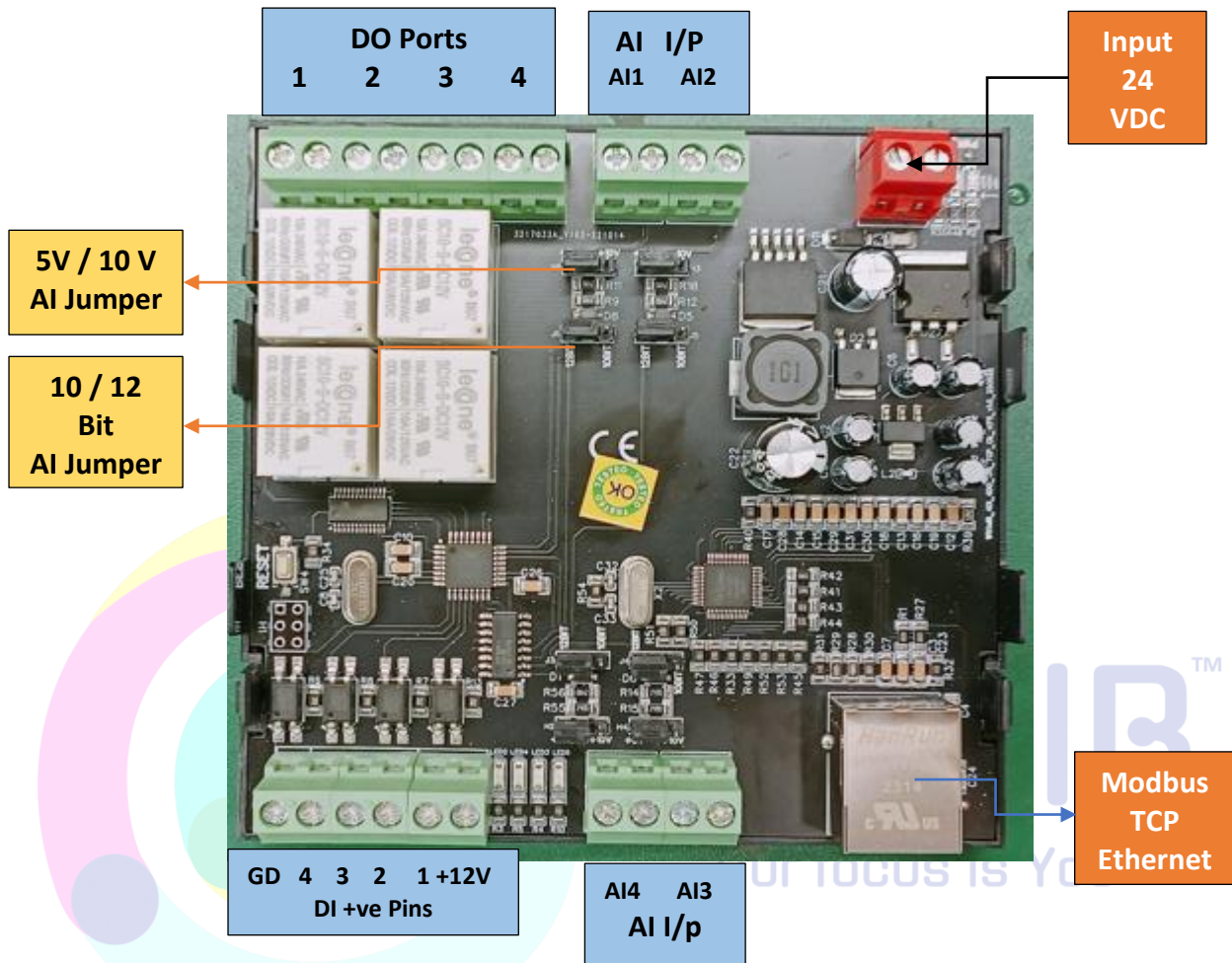
Note: Ensure that the connected load does not exceed the relay's contact rating of **10A @ 250VAC** (or **10A @ 125VAC**) or **10A @ 30VDC** (or **10A @ 28VDC**).

AI



- **Analog Input (AI) Operation:** Each channel provides two terminals: + (Positive) and - (Negative/Common). These are used to read external analog sensor signals.
- **Channel Configuration (Visual Guide):** Each of the four AI channels features two dedicated jumpers to independently set the voltage range and bit resolution. Refer to the color-coded boxes in the board diagram to configure your desired scenario:
 - **Red Box (5V, 10-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **10 BIT**.
 - **Green Box (5V, 12-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **12 BIT**.
 - **Blue Box (10V, 10-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **10 BIT**.
 - **Yellow Box (10V, 12-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **12 BIT**.
- **Sensor Wiring Connections:**
 - The analog sensor's positive signal wire should be connected to the **+** terminal of the desired channel.
 - The sensor's common or ground wire should be connected to the corresponding **-** terminal.
- **Crucial Setup Recommendation:** Always ensure that the voltage jumper setting strictly matches the maximum output range of your connected sensor (either 5V or 10V) to guarantee accurate analog-to-digital conversion and prevent data clipping.

Product Image:



Note – Configuration Guide is separately provided for the configuration of the module.

For more details: <http://www.youtube.com/@wittelb>

Contact Information

Augmatic Technologies Pvt. Ltd.,
Plot no 6, Shah Industrial Estate II,
Kotambi,
Vadodara – 391510.

Email – Sales@wittelb.com

Support: support@wittelb.raiseaticket.com / +91 94038 92805